Signal Check Device

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Abstract

Epilepsy alarms can detect epileptic seizures and alert healthcare staff, but wireless communication can be affected by distance and different building materials. This can make it difficult to identify a suitable location for the control unit and ensure reliable communication. To address this problem, a low-cost signal strength indicator was developed using an ESP32-C3 and three LEDs to represent different RSSI levels at 2.4 GHz. The prototype successfully detects and classifies signal strength as strong, medium or weak, demonstrating that a simple and cost-effective signal strength indicator can be developed using affordable components.

1. Introduction

Epilepsy is a neurological disease that causes epileptic seizures due to overactivation of nerve cells in the brain [1]. Wearable non-invasive devices can detect epileptic seizures through measurements of movements [2]. EpiSafe is a wrist-worn sensor that measures arm movements in three different directions [3]. The sensor sends alarm signals to the control unit [3]. According to the manufacturer, the range between the wristband sensor and control unit is up to 100 meters with an unobstructed view [3]. However, the range can be affected by different building materials [3]. Received Signal Strength Indicator (RSSI) is a measurement that represents the signal power on a radio link [4]. RSSI is affected by for example distance, physical obstacles and different construction materials [4].

The aim of this project is to develop a signal strength indicator that helps identify a suitable placement for the EpiSafe control unit, for example in a nursing home. The indicator uses three colours to represent different signal-quality levels. The prototype is designed as a simple and cost effective way to demonstrate the feasibility of a signal strength indicator for this purpose. It will be tested at 2.4 GHz over different distances and through different wall materials. The 2.4 GHz band was chosen as a practical prototype platform because it is licence-exempt in Sweden while the 869.2–869.25 MHz band is designated for social-alarm devices and is subject to specific technical requirements [6]. Since radio propagation is frequency-dependent, the measurements at 2.4 GHz are intended to evaluate the functionality of the signal indicator rather than reproduce the exact performance of the EpiSafe system.

2. Design Section

The prototype consists of an ESP32-C3 board connected to three LEDs, which provide a visual indication of received Wi-Fi signal strength. A TP-Link TL-WR802N wireless router serves as the signal source during testing, as shown in figure 1. The ESP32-C3 scans for a predefined Wi-Fi network and retrieves its Received Signal Strength Indicator (RSSI), expressed in dBm. The measured value is compared against predefined thresholds, and the corresponding LED is activated to indicate the signal-strength classification. Green indicates a high signal level, yellow indicates a medium signal level, and red indicates either a low signal level or that the target network was not detected. The RSSI value and network detection status are also printed to the serial monitor for testing and debugging.

For the prototype's preliminary classification, RSSI values of −67 dBm or higher were classified as high, values below −67 dBm and down to −75 dBm as medium, and values below −75 dBm as low. These thresholds were informed by recommended 2.4 GHz Wi-Fi RSSI benchmarks [8, 9]. The implementation further distinguishes between low signal strength (down to −83 dBm) and zero or no signal (below −83 dBm or when the target network is not detected). Both classifications are indicated by the red LED. These thresholds are used to demonstrate the indicator's functionality and are not validated acceptance limits for the EpiSafe system.

The prototype was programmed in C++ using the Arduino IDE and the ESP32 Wi-Fi library. Measurements are repeated at regular intervals, allowing the user to observe changes in signal strength when moving the device between locations. The receiver, LEDs and associated wiring are assembled on a breadboard and housed in a custom 3D-printed enclosure. This setup provides a portable proof of concept for visually assessing Wi-Fi signal strength in different environments.

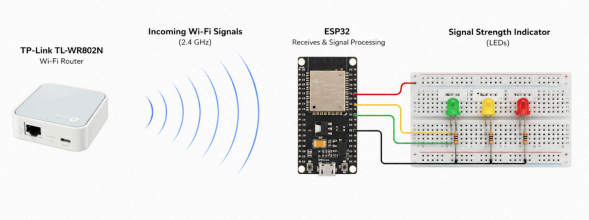


Figure 1: AI-generated (ChatGPT, 2026) image showing the basic system overview of the prototype.

3. Hypothesis and Expected Results

It was hypothesised that the received signal strength indicator (RSSI) would decrease as the distance between the Wi-Fi router and receiver increased. Consequently, the LED classification was expected to change from high to medium and eventually to low as the distance increased. To evaluate the behaviour of the signal strength indicator, seven trials were conducted to record the distance at which the device changed classification at the predefined RSSI thresholds of −67 dBm and −75 dBm. The measurements were performed without physical obstructions between the router and receiver.

Average distance (m)	Trials (1-7)
-67 dBm	10.26
-75 dBm	22.96

Table 1: Data of the average distance between source and prototype to where the corresponding dBm readings were collected.

The transition from high to medium signal strength occurred at distances ranging from 9.1 to 11.9 metres, with a mean distance of approximately 10.3 metres across the seven trials. The transition from medium to low signal strength occurred at distances ranging from 20.1 to 24.5 metres, with a mean distance of approximately 23.0 metres. The results demonstrate that the device can indicate changes in received Wi-Fi signal strength through its LED classifications. However, the variation in transition distances between trials indicates that the relationship between distance and RSSI is not perfectly consistent, even under unobstructed conditions. These findings apply to the specific router and receiver configuration used in the experiments. Further testing would be required to evaluate the effects of physical obstacles, different building materials and interference from other devices operating in the 2.4 GHz band.

4. Conclusion

The prototype demonstrated the basic functionality of a signal strength indicator using an ESP32-C3 and three LEDs. The device scans for a predefined Wi-Fi network, measures its received signal strength (RSSI), and displays the signal level using green, yellow and red LEDs. In seven trials without physical obstructions, the transition from high to medium signal strength occurred at a mean distance of approximately 10.3 metres, while the transition from medium to low occurred at approximately 23.0 metres. These results demonstrate the indicator's functionality under the tested conditions but do not directly represent the coverage of the EpiSafe system, which operates at a different frequency.Future improvements could include a switching mechanism to detect signals at different frequencies and a touchscreen interface with a keyboard to enter network names or select signal sources.



Figure 2: Photo showing the completed prototype while displaying high RSSI.

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Appendix Section A

Raw test data presented as mean/averages in Table 1.

Distance (m)	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Trial 6	Trial 7
-67 dBm	10.2	9.8	10.1	10.3	10.4	9.1	11.9
-75 dBm	23.9	23.8	24.5	24.2	23.9	20.3	20.1

Appendix Section B

Source code as well as additional documentation of the project is available via the following Github link: https://github.com/FrejStenberg/CM2024-RF-Signal-Tester